A Tale of Five Decoders.

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Introduction

Today:

- Recap, and were we stood before the summer.
- Simulations Results.
- Where we heading.

Map

Simulations

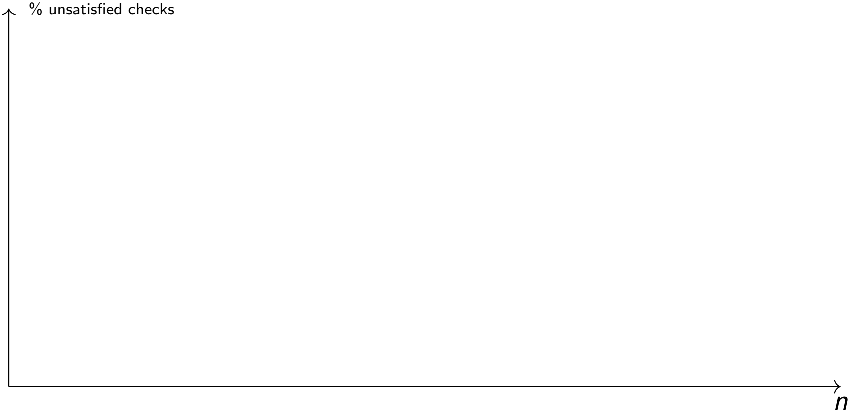
Perskil and Swift Rule?

Dennis at el?

Is the (d, k)-Toric a MEM?

Coming with $O(1)$-depth
gates for 'good' codes.

How to Read



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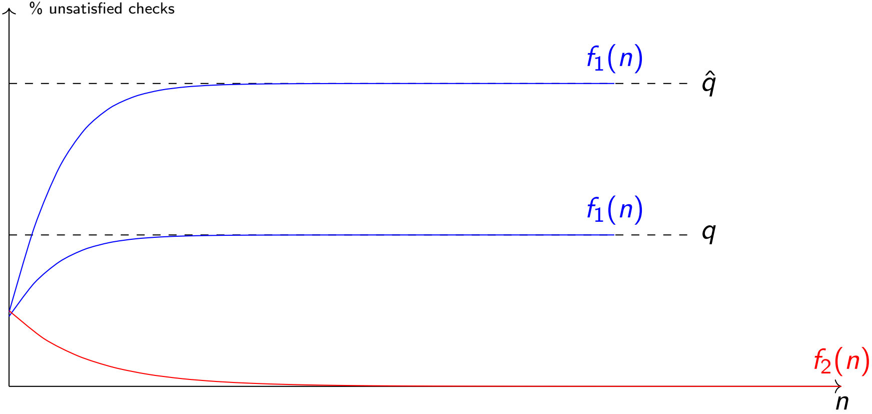
error rates :[0.001]
error accumulation :False

Ideal Result.



colors

In red, No accumulation of errors. In blue decoding in the presence of noise.

Majority

Alg.

Each bit looks at its neighbour checks, if the majority of them is unsatisfied, then it flip itself.

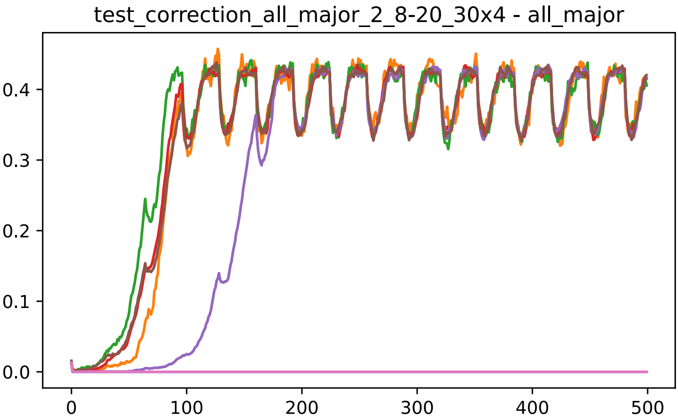
Justification.

Was suggested in Dennis.

Bottom line.

- Fails to correct, even in the absence of noise.
- Collisions.
- Gives better result when requiring $(\frac{1}{2} + \mu)$-majority.

Majority



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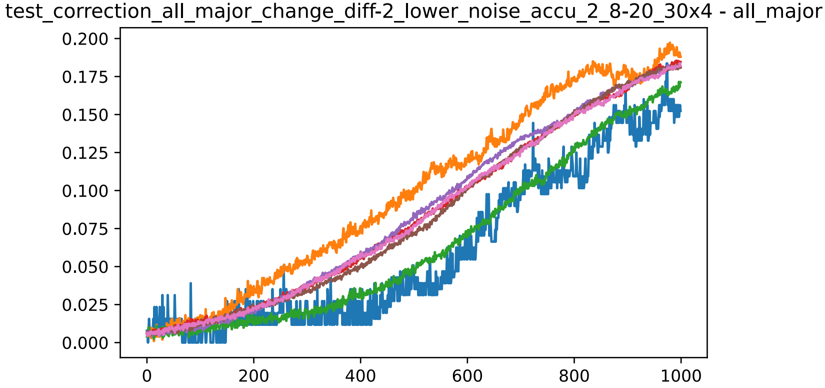
error accumulation :False

Majority



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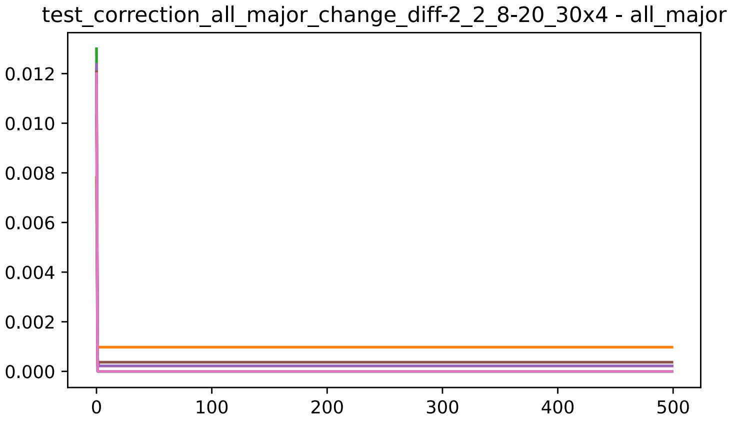
Majority



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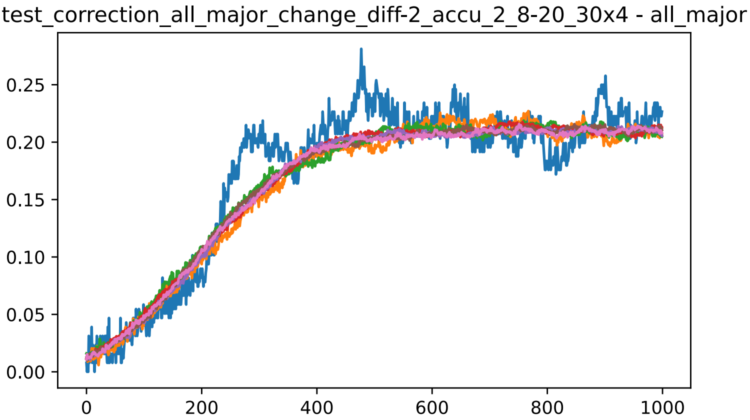
error accumulation :False

Majority



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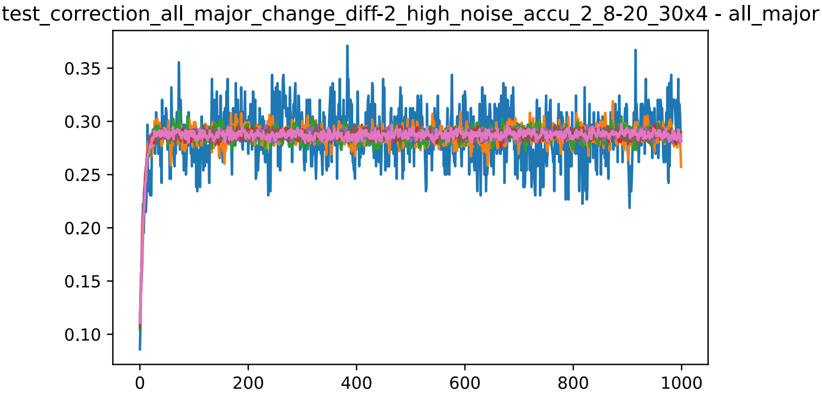
Majority



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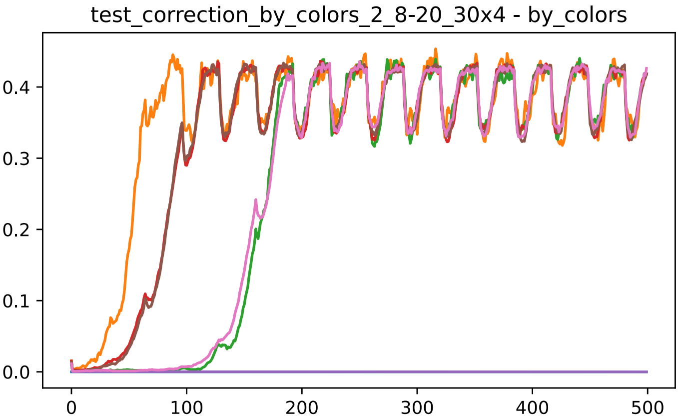
error accumulation :True

Colors



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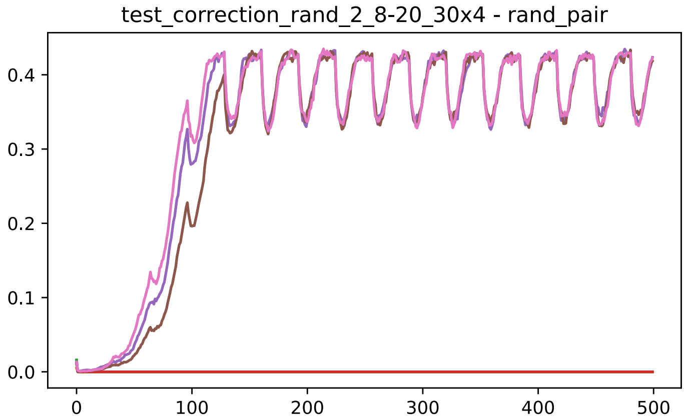
error accumulation :False

Picking Random



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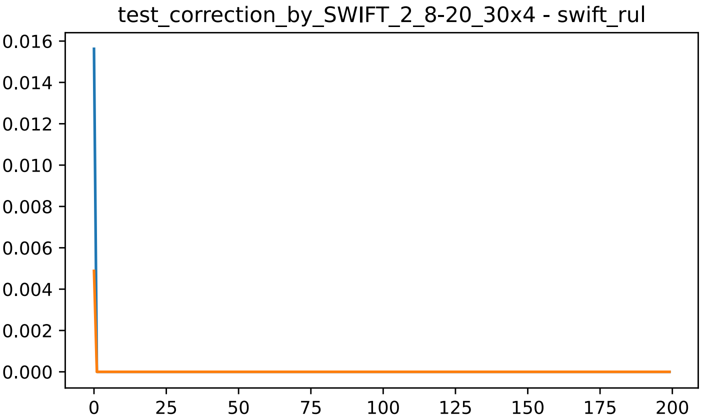
error accumulation :False

SWIFT



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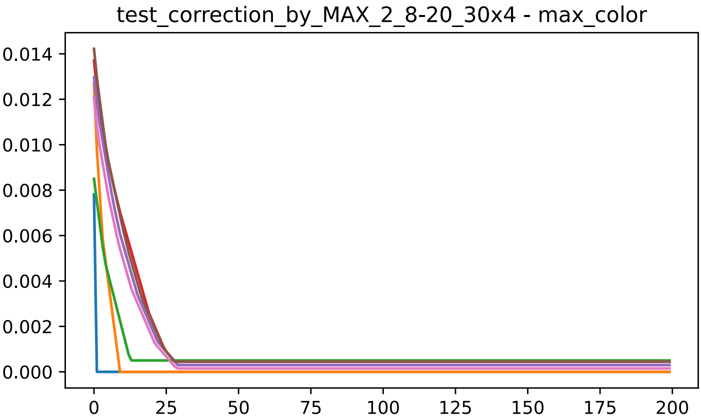
error accumulation :False

Max Color



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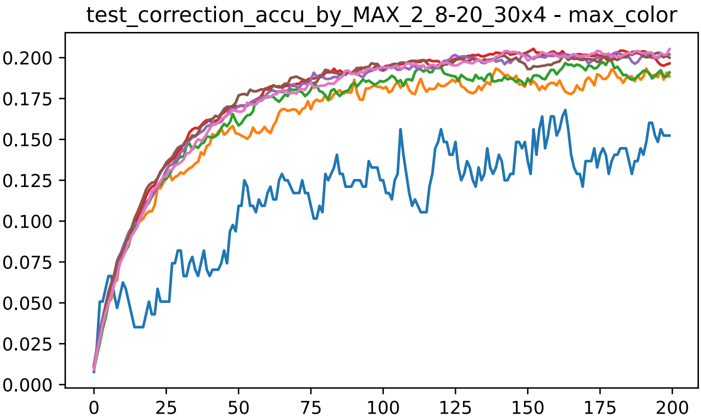
error accumulation :False

Max Color



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error rates :[0.001]
error accumulation :True